

EXPERIMENT 7

PROGRAM:

```
clc; clear; close all
```

```
fs=1000; t=(0:999)/fs;  
s=sin(2*pi*100*t)+0.5*sin(2*pi*200*t);  
n=0.3*randn(size(t));  
x=s+n;
```

```
figure, plot(x), grid on
```

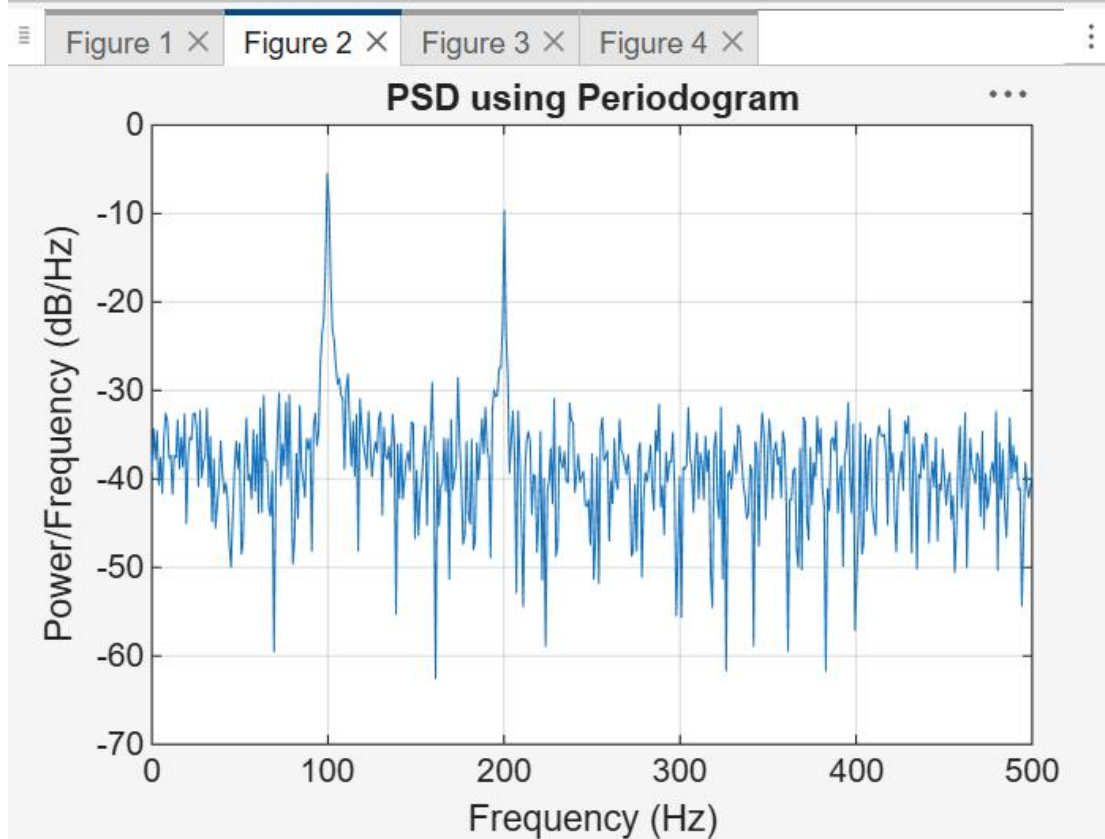
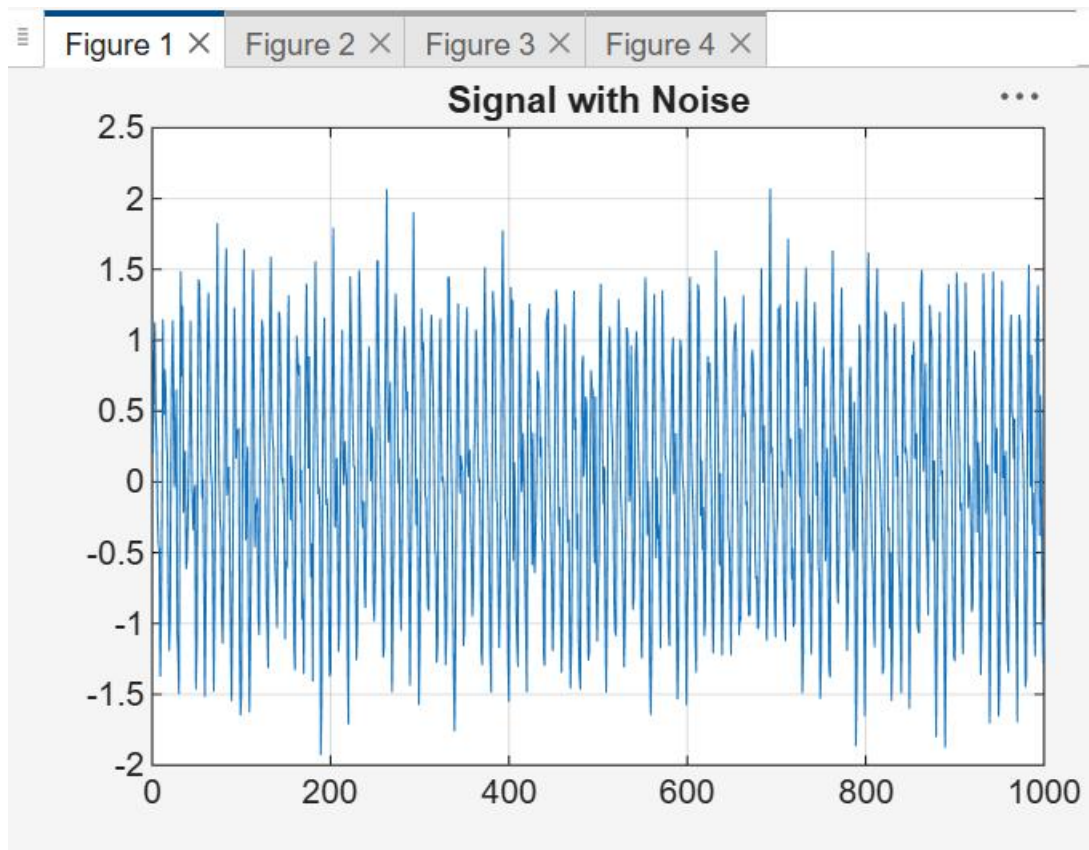
```
title('Signal with Noise')
```

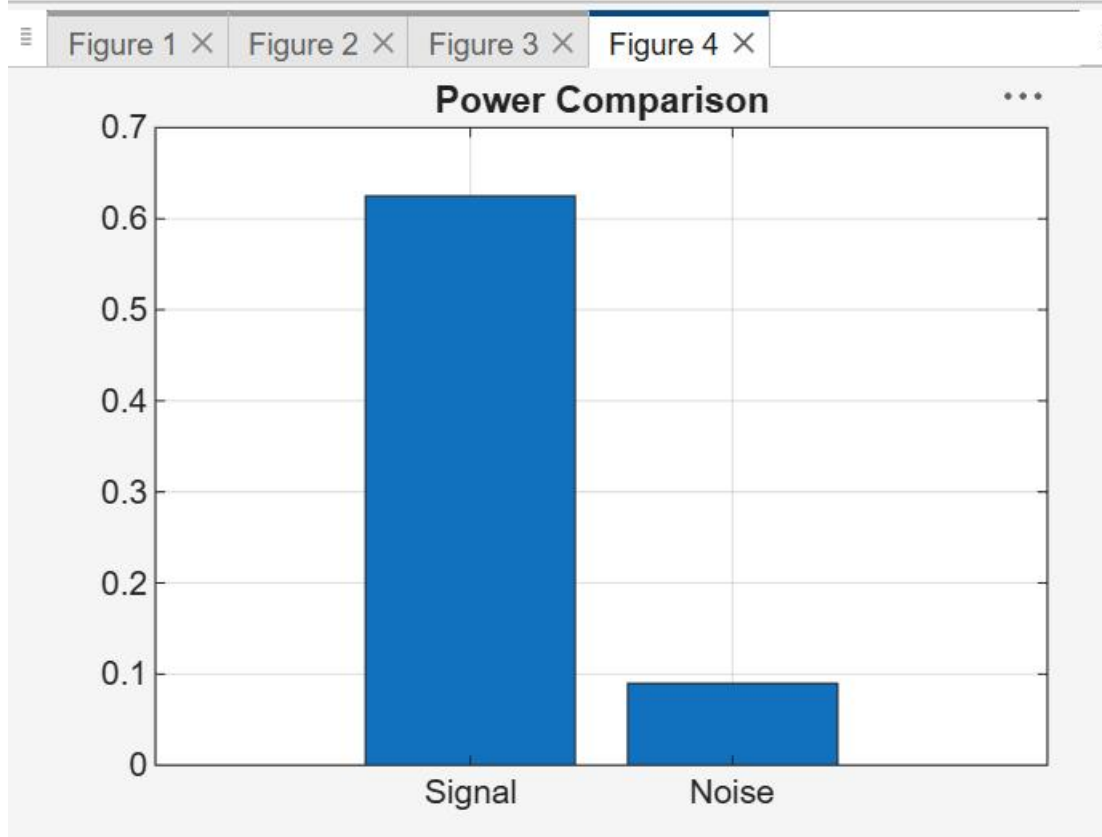
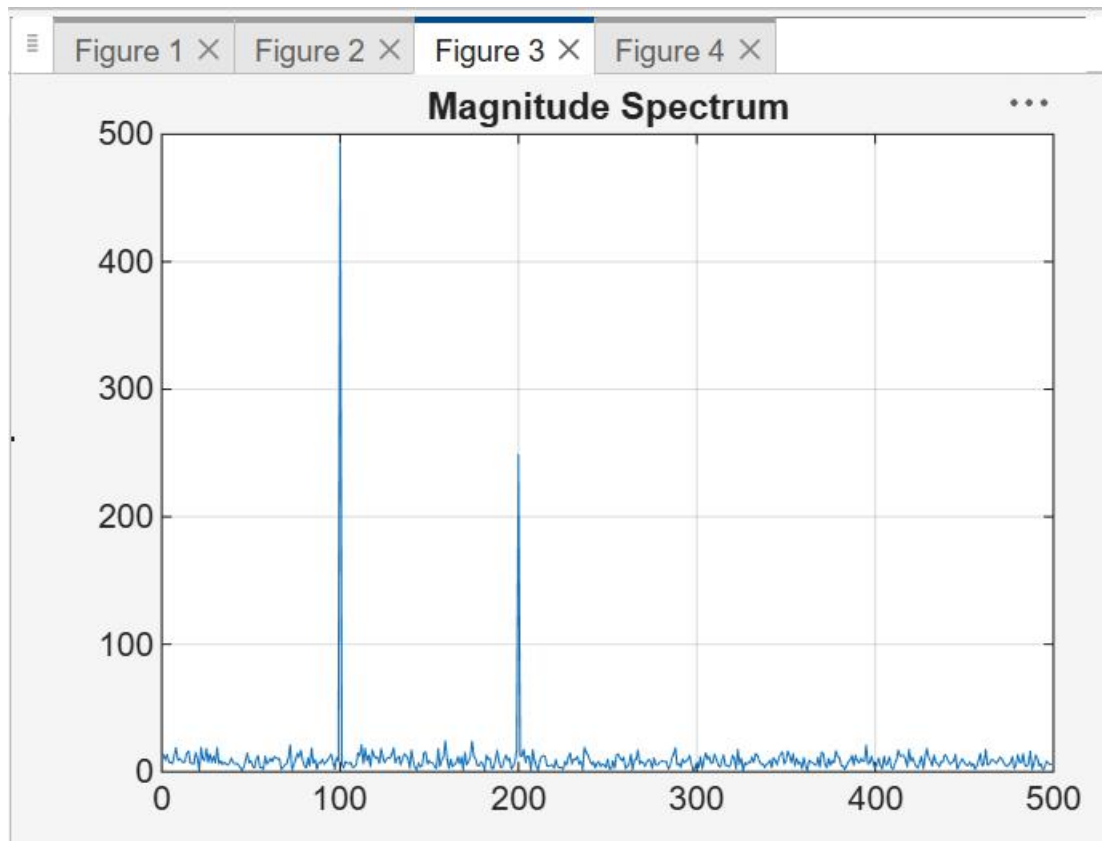
```
figure, periodogram(x,[],[],fs)  
title('PSD using Periodogram')
```

```
figure  
X=abs(fft(x));  
f=(0:length(X)-1)*fs/length(X);  
plot(f(1:500),X(1:500)), grid on  
title('Magnitude Spectrum')
```

```
figure  
bar([mean(s.^2) var(n)])  
set(gca,'XTickLabel',{'Signal','Noise'})  
title('Power Comparison')  
grid on
```

RESULT:





EXP7.2 PROGRAM

```
clc; clear; close all;
```

```
% Signal
```

```
fs = 1000; N = 256; t = (0:N-1)/fs;
```

```
x = sin(2*pi*100*t) + 0.8*sin(2*pi*110*t);
```

```
NFFT = 2048; f = (0:NFFT/2-1)*(fs/NFFT);
```

```
% Windows
```

```
w = {rectwin(N), hamming(N), hann(N)};
```

```
names = {'Rectangular','Hamming','Hanning'};
```

```
% Processing
```

```
for i = 1:3
```

```
    xw = x(:).*w{i};
```

```
    X = fft(xw,NFFT);
```

```
    P{i} = abs(X(1:NFFT/2));
```

```
    P{i} = P{i}/max(P{i});
```

```
end
```

```
%% Individual Spectrums
```

```
figure;
```

```
for i = 1:3
```

```
    subplot(3,1,i);
```

```
    plot(f,20*log10(P{i}), 'LineWidth',1.5);
```

```
    title(['Spectrum using ' names{i} ' Window']);
```

```
    xlim([0 250]); grid on;
```

```
end
```

```
%% Comparison
```

```
figure;
```

```
plot(f,20*log10(P{1}), 'k', f,20*log10(P{2}), 'r', f,20*log10(P{3}), 'b', 'LineWidth',1.5);
```

```
legend(names); title('Window Comparison'); xlim([50 170]); grid on;
```

```
%% Window Shapes
```

```
figure;
```

```
for i = 1:3
```

```
    subplot(3,1,i); plot(w{i}, 'LineWidth',1.5);
```

```
    title([names{i} ' Window']); grid on;
```

```
end
```

```
%% Peak Detection
```

```
for i = 1:3
```

```
    [pks{i},loc{i}] = findpeaks(P{i},f, 'MinPeakHeight',0.3);
```

```
    fprintf('\n%s Window Frequencies:\n',names{i});
```

```
disp(loc{i});
end
```

```
%% Peak Marking (Hamming)
figure;
plot(f,20*log10(P{2}),'LineWidth',1.5); hold on;
plot(loc{2},20*log10(pks{2}),'ro','MarkerFaceColor','r');
title('Detected Frequencies (Hamming)'); xlim([50 170]); grid on;
```

```
%% Main Peaks
fprintf('\nMain Spectral Peaks:\n');
for i = 1:3
    [~,idx] = max(P{i});
    fprintf('%s Window Peak = %.2f Hz\n',names{i},f(idx));
end
```

Output:

Rectangular Window Frequencies:

100.0977 109.8633

Hamming Window Frequencies:

100.0977 109.8633

Hanning Window Frequencies:

100.0977 109.8633

Main Spectral Peaks:

Rectangular Window Peak = 100.10 Hz

Hamming Window Peak = 100.10 Hz

Hanning Window Peak = 100.10 Hz

RESULT:

